

PROJECT REPORT

Trip Budget Predictor

(Java Console Application)

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This report explains a small Java program made to help with a common travel-planning task: getting a rough idea of how much a trip may cost. The language has been kept simple so that the report reads like a student's own project explanation rather than a highly technical document.

1. Abstract

The Trip Budget Predictor is a simple Java console program that helps a traveller work out an estimated budget before going on a trip. The user enters details such as the destination, number of people, number of days, travel distance, food cost, hotel cost, activity cost and other expenses. The program then adds these amounts and shows the total expected budget along with the approximate amount for one person.

The main purpose of making this project was to practise basic Java programming in a practical situation. It uses keyboard input, variables, methods, arithmetic calculations, if-else conditions and formatted output. The program is small, but it shows how these basic concepts can be combined to solve a real-life calculation problem.

2. Introduction

2.1 Background

When people plan a trip, they usually have to think about several expenses at the same time. Travel, hotel, food and sightseeing are common examples. If the number of people or the number of days changes, the calculations also have to be changed. Doing all of this by hand can take time and can lead to small mistakes.

This project was made as a simple solution to that problem. Instead of calculating every amount separately, the user enters the required information once and the Java program does the basic calculations automatically.

2.2 Objectives

- 1 Make a simple Java program for estimating the cost of a trip.
- 2 Take trip information from the user through the console.
- 3 Calculate travel, food, hotel, activity and miscellaneous expenses.
- 4 Show the total estimated budget and the approximate cost per person.
- 5 Put the trip into a budget, moderate or premium category based on the calculated per-person amount.

3. System Analysis

3.1 Existing Method

- Manual calculation: A traveller may calculate each expense separately using a calculator.
- Spreadsheet method: A spreadsheet can be useful, but the formulas and values still have to be entered and maintained.
- Chance of mistakes: If the number of travellers or days changes, several calculations may need to be changed again.

3.2 Proposed System

The proposed system is a small Java-based budgeting tool. The user gives the required trip details, and the program uses separate methods to work out each part of the budget. At the end, it prints a short summary so the user can see where the money is going and what the estimated cost per person will be.

Main benefits:

- It performs several calculations quickly.
- It has a straightforward console interface.
- The calculations are divided into separate methods, which makes the program easier to follow.
- It gives an immediate per-person estimate.

4. System Design

4.1 Basic Structure

- Input: Scanner is used to collect information from the keyboard.
- Processing: Java methods perform the calculations for the different expense categories.
- Output: The final result is printed in a readable format on the console.

4.2 How the Data Moves Through the Program

1. The user enters the destination, number of people, number of days, distance and estimated expenses.
2. The program applies the required formula to each expense.
3. All the calculated amounts are added together.
4. The total and per-person amount are displayed, along with the budget category.

4.3 Main Formulae

Expense	Calculation
Transport	Round-trip distance × estimated cost per kilometre
Food	Days × Persons × Daily food cost per person
Hotel	Days × Persons × Daily hotel cost per person
Activities	Days × Persons × Daily activity cost per person

Expense	Calculation
Total Budget	Transport + Food + Hotel + Activities + Miscellaneous
Cost Per Person	Total Budget ÷ Number of Persons

5. Implementation Details

5.1 Hardware and Software Requirements

Requirement	Details
Operating System	Windows / macOS / Linux
Programming Language	Java
Java Version	JDK 8 or later
Editor / IDE	VS Code / IntelliJ IDEA / Eclipse / NetBeans
RAM	2 GB or more recommended

5.2 Java Concepts Used

- Scanner – takes input from the user through the keyboard.
- Methods – keeps different calculations in separate reusable functions.
- Variables – stores values such as destination, people, days, distance and expenses.
- Arithmetic operators – handles multiplication, addition and division.
- if-else – decides the budget category from the per-person cost.
- printf – prints money values with two decimal places.

5.3 Main Functions

Function	Purpose
calculateTransport()	Works out the round-trip transport expense.
calculateFood()	Calculates food cost for the full trip.
calculateHotel()	Calculates accommodation cost.
calculateActivities()	Calculates sightseeing and activity cost.
main()	Takes input, calls the methods and prints the final result.

5.4 Sample Input

Field	Sample value
Destination	Manali
Persons	4
Duration	5 days
One-way distance	600 km
Daily food/person	Rs 500
Daily hotel/person	Rs 900
Daily activities/person	Rs 400
Miscellaneous	Rs 2,000

For this example, transport is calculated using the round-trip distance and an estimated rate of Rs 12 per kilometre. The other amounts use the daily values entered above.

6. Features and Functionality

6.1 User Input

The program asks for the main information needed to estimate the trip cost. This keeps the calculation based on the user's own plan.

6.2 Automatic Calculation

After the values are entered, the separate Java methods calculate each expense instead of making the user do the arithmetic manually.

6.3 Per-Person Cost

The total estimated budget is divided by the number of travellers. This gives an approximate amount that each person would need to contribute.

6.4 Budget Classification

The program uses fixed limits to give a simple category to the trip.

Per-person amount	Category
Up to Rs 5,000	LOW / BUDGET TRIP
Rs 5,001 – 10,000	MODERATE TRIP
Above Rs 10,000	PREMIUM TRIP

7. Testing and Expected Output

The program can be checked by changing values such as the number of travellers, trip length and daily expenses. The following example uses the sample values given earlier.

Item	Calculation	Amount
Transport	$600 \times 2 \times \text{Rs } 12$	Rs 14,400
Food	$5 \times 4 \times \text{Rs } 500$	Rs 10,000
Hotel	$5 \times 4 \times \text{Rs } 900$	Rs 18,000
Activities	$5 \times 4 \times \text{Rs } 400$	Rs 8,000
Miscellaneous	Given value	Rs 2,000
Total	Sum of all expenses	Rs 52,400
Per person	$\text{Rs } 52,400 \div 4$	Rs 13,100

With a calculated cost of Rs 13,100 per person, this particular example falls in the PREMIUM TRIP category according to the fixed limits used by the program.

8. Advantages and Limitations

8.1 Advantages

- Easy to use for a basic trip estimate.
- Saves time compared with repeating the same calculations by hand.
- The use of separate methods makes the code easier to read and maintain.
- Shows both the total amount and the estimated amount per person.

8.2 Limitations

- The transport calculation uses a fixed estimated rate and does not check live ticket prices.
- Food, hotel and activity costs depend on the estimates entered by the user.
- The current version works only through the console and does not save old trips.
- Taxes, discounts, currency conversion and live booking information are not included.

9. Future Scope

- 1 GUI: Add a Java Swing or JavaFX interface so the program is easier to use.
- 2 Trip History: Save previous calculations in a file or database.
- 3 Multiple Transport Modes: Keep separate calculations for train, bus, car and flight.
- 4 Dynamic Pricing: Let the user enter different transport and hotel rates.
- 5 Currency Support: Allow the estimated budget to be viewed in another currency.
- 6 Budget Alerts: Show a warning when the estimated cost goes above a chosen limit.

10. Conclusion

This project shows how basic Java concepts can be used to solve a simple everyday problem. The program takes travel details from the user, performs the required calculations and gives an estimate of the overall trip cost and the amount per person. While it is not meant to replace real-time travel prices, it is a useful starting point for learning how input, methods, calculations and conditions work together in a Java application.

11. References

- 1 Oracle Java Documentation – Java Platform Documentation.
- 2 Java Scanner and input/output concepts – standard Java documentation.
- 3 Herbert Schildt, Java: The Complete Reference.
- 4 Project calculations and design prepared for the Trip Budget Predictor.

Project Deliverable

The Java source file for the project is named TripBudgetPredictor.java. It can be compiled and run with a Java JDK.